



Tank Inspection Report
for
Enva
Tank CT-3
at
Shanbally, Co. Cork, Cork
Inspection date 23rd July 2018

Author	Admin Chk 1	Admin Chk 2	Date
CH	FF	MD	23.07.2018





TANK INSPECTION REPORT

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TANK INSPECTION REPORT

CLIENT:	Enva
CONTRACTOR:	MTI
INSPECTION LOCATION	Shanbally, Co. Cork
REPORT NUMBER:	5675.10
INSPECTION DATE:	23rd July 2018
SUBJECT:	Tank CT-3
SCOPE OF INSPECTION	Visual Inspection , Ultrasonic Thickness Measurements,
EQUIPMENT:	Digital Camera, DM5 UT Meter ,Mirror, Pit Gauge
CONSUMABLES:	UCA-2M Ultrasonic Couplant
NDE SPECIFICATION:	

Introduction

MTI was requested to carry out a periodic inspection of Tank No. CT-3 at Shanbally, Co. Cork. The purpose of the inspection was to provide inspection and NDT data for determination of fitness for service by others.

The agreed workscope was;

External visual inspection of vessel and attachments

Readings at 0.5m intervals around shell, on each shell course where accessible

Four readings per nozzle and comp pad.

Ten readings on each roof plate

Extra readings in areas where corrosion is noted

Background

Enva operates a water treatment facility at Shanbally, Co. Cork..



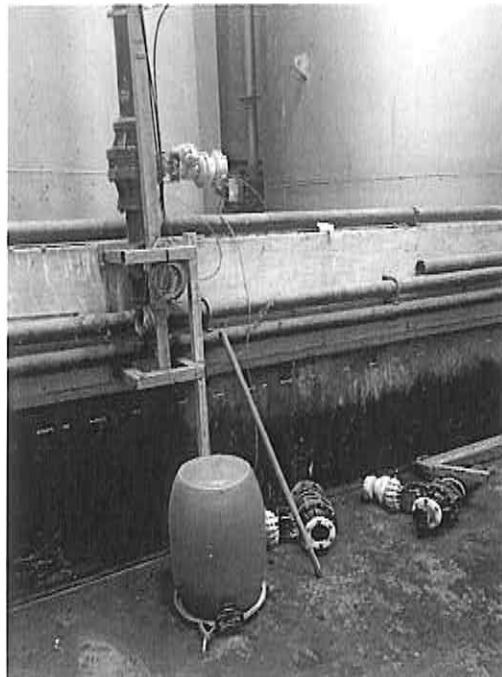
Visual inspection

External visual inspection of the following items was carried out to locate signs of corrosion, leaks or mechanical damage;

- All shell course welds where accessible
- Shell where accessible
- Shell attachments / nozzles

Findings

- Corrosion / Pitting band on the bottom of the shell around its perimeter
- approximately 50 mm wide and extending into the outer base rim, random areas were selected and the corrosion was removed, a pitting depth of 1.0 mm to 2.0 mm was measured at these locations.
- All of this corrosion would need to be removed to properly assess the depth of the overall pitting present.
- The handrails and platform appear to be in good condition.
- Note: No safe access to the top section of the shell.
- Ultrasonic thickness readings taken on the first 2 courses only,
- Note: Shell nozzles are numbered clockwise starting at the man way
- Debris litter in bund and surrounding areas





ULTRASONIC THICKNESS DETERMINATION REPORT

FORM NUMBER: UTMREP/001/GP

CLIENT	ENVA Ringaskiddy		
LOCATION	ENVA Shanbally, Co. Cork		
REPORT No	5675.10.02	CLIENT REQ. No.	N/A
CLIENT JOB NO AND DESCRIPTION	Tank CT-3		
TEST DATE(S)	23rd July 2018		
SPECIFICATION	ASME BPVC Section V Article 5	MTI SOP	4.1 Rev 03
WORKSCOPE	Spot-check Ultrasonic thickness measurement of tank nozzle bodies.		
EQUIPMENT	GE DM5E s/n DM5EG1502346 thickness meter with DA-312 probe		
COUPLANT	Sonagel		
TECHNIQUE	MTI/UTM/001		

Ultrasonic Thickness Measurements taken on all second course and roof nozzles,
Note: No construction drawing available.

Level	1	2	3	4	5	6	7	8
Shell Level 1	6.0	6.0	6.1	6.0	6.0	6.1	6.0	6.1
Shell Level 2	6.0	6.2	6.3	6.1	6.1	6.0	6.0	6.0
Roof Plate 1	5.9	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Roof Plate 2	6.0	5.9	6.1	6.0	6.1	6.1	5.9	6.0

SHELL NOZZLE THICKNESS READINGS

Nozzle	Diameter	12 o clock	3 o clock	6 o clock	9 o clock
1	24"	6.0	6.0	5.9	6.0
2	2"	5.4	5.2	5.1	5.2
3	2"	3.8	3.7	3.9	3.9
4	4"	6.0	6.3	6.4	6.3
5	3"	5.4	5.3	5.4	5.2





MATERIAL TEST AND INSPECTION LTD.

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HOLLIDAY DETECTION INSPECTION REPORT

FORM NUMBER: HDREP/001GP

CLIENT:	ENVA IRELAND LTD.
LOCATION:	Ringaskiddy, Co Cork
REPORT NUMBER:	5675.16
SERVICE	Aluminium Chloride
CAPACITY	50000 Litres
MATERIAL	G R P
TEST DATE:	23.07.18
TANK NO.	CT 5
SCOPE:	A. Visual inspection. B. Holiday detection externally on the vessel where accessible.
ACCEPTANCE CRITERIA:	Report all defects
EQUIPMENT & VOLTAGE USED:	Maple holiday detector 5 kV
TEST RESULTS:	A. No bulging, mechanical damage or other defects noted. Product leakage on top of vessel from nozzles.no safe access for further investigation. B. The fibreglass appears to be intact, and no defect alarms were located at the time of this inspection

TECHNICIAN  Conor Hudson CSWIP 3.0	SIGNED FOR MTI LTD  <i>Material Test & Inspection Ltd.</i>	CLIENT ACCEPTANCE	INSPECTION AUTHORITY APPROVAL
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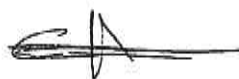
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HOLLIDAY DETECTION INSPECTION REPORT

FORM NUMBER: HDREP/001GP

CLIENT:	ENVA IRELAND LTD.
LOCATION:	Ringaskiddy, Co Cork
REPORT NUMBER:	5675.17
SERVICE	Aluminium Chloride
CAPACITY	50000 Litres
MATERIAL	G R P
TEST DATE:	23.07.18
TANK NO.	CT 6
SCOPE:	A. Visual inspection. B. Holiday detection externally on the vessel where accessible.
ACCEPTANCE CRITERIA:	Report all defects
EQUIPMENT & VOLTAGE USED:	Maple holiday detector 5 kV
TEST RESULTS:	A. No bulging, mechanical damage or other defects noted. Product leakage on top of vessel from nozzles. no safe access for further investigation. B. The fibreglass appears to be intact, and no defect alarms were located at the time of this inspection

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HOLLIDAY DETECTION INSPECTION REPORT

FORM NUMBER: HDREP/001GP

CLIENT:	ENVA IRELAND LTD.
LOCATION:	Ringaskiddy, Co Cork
REPORT NUMBER:	5675.18
SERVICE	Aluminium Chloride
CAPACITY	50000 Litres
MATERIAL	G R P
TEST DATE:	23.07.18
TANK NO.	CT 7
SCOPE:	A. Visual inspection. B. Holiday detection externally on the vessel where accessible.
ACCEPTANCE CRITERIA:	Report all defects
EQUIPMENT & VOLTAGE USED:	Maple holiday detector 5 kV
TEST RESULTS:	A. No bulging, mechanical damage or other defects noted. Product leakage on top of vessel from nozzles. no safe access for further investigation. B. The fibreglass appears to be intact, and no defect alarms were located at the time of this inspection

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